C1 extends C2

```java
package javacore;


class C2 {
    int a = 10;


    C2() {
        System.out.println("C2 constructor called");
    }


    void displayC2() {
        System.out.println("Value of a = " + a);
    }
}


class C1 extends C2 {
    int b = 20;


    C1() {
        super();
        System.out.println("C1 constructor called");
    }


    void displayC1() {
        System.out.println("Value of b = " + b);
        System.out.println("Sum = " + (a + b));
    }
}
```

```
public class Program1 {
   public static void main(String[] args) {

       C1 c = new C1();

       c.displayC2();
       c.displayC1();
   }
}
```

Execution Steps

1. Start the program.

2. The main() method starts execution.

3. A C1 object is created:

   C1 c = new C1();

4. The C1 constructor is called.

5. Inside the C1 constructor, the statement

   super();

   calls the constructor of the parent class C2.

6. C2 constructor executes and displays:

   C2 constructor called

7. Control returns to the C1 constructor.

8. C1 constructor displays:

    C1 constructor called

9. The statement

    c.displayC2();

  calls the displayC2() method of C2.

10. It displays the value of a:

    Value of a = 10

11. The statement

    c.displayC1();

  calls the displayC1() method of C1.

12. It displays the value of b:

    Value of b = 20

13. It calculates a + b:

    10 + 20 = 30

14. It displays:

    Sum = 30

15. Program execution ends.

Output

C2 constructor called

C1 constructor called

Value of a = 10

Value of b = 20

Sum = 30

C1 Implements I1

```
package javacore;


interface I1 {
   int a = 90;


   void mul();
}


class C1 implements I1 {


   int b = 2, c = 3;


   @Override
   public void mul() {
      System.out.println("Mul is " + (b * c));
   }


   public static void main(String[] args) {


      C1 c = new C1();


      c.mul();
      System.out.println("Value of a is " + c.a);
   }
```

}

Execution Steps

1. Start the program.

2. The main() method starts execution.

3. An object of C1 class is created:

```
C1 c = new C1();
```

4. C1 implements the I1 interface.

   Therefore, C1 must provide the mul() method.

5. The statement

```
c.mul();
```

   calls the mul() method of C1.

6. Inside mul():

```
b = 2
c = 3
```

   Multiplication is calculated:

```
b * c = 2 * 3 = 6
```

7. It displays:

```
Mul is 6
```

8. The statement

```
System.out.println("Value of a is " + c.a);
```

accesses variable a from the I1 interface.

9. The value of a is:

   a = 90

10. It displays:

   Value of a is 90

11. Program execution ends.

Output

Mul is 6

Value of a is 90

C1 Implements I1,I2

```
package javacore;


interface I1 {
   int a = 90;


   void mul();
}


interface I2 {
   int x = 10;


   void add();
}
```

```java
class C1 implements I1, I2 {

    int b = 2;
    int c = 3;

    @Override
    public void mul() {
        System.out.println("Multiplication = " + (b * c));
    }

    @Override
    public void add() {
        System.out.println("Addition = " + (b + c));
    }

    public static void main(String[] args) {

        C1 obj = new C1();

        obj.mul();
        obj.add();

        System.out.println("Value of a = " + obj.a);
        System.out.println("Value of x = " + obj.x);
    }
}
```

Execution Steps

1. Start the program.

2. The main() method starts execution.

3. An object of C1 class is created:

    C1 obj = new C1();

4. C1 implements two interfaces:

    I1 and I2

5. Interface I1 contains:

    a = 90

    mul()

6. Interface I2 contains:

    x = 10

    add()

7. C1 provides implementations for both methods:

    mul()

    add()

8. The statement

    obj.mul();

  calls the mul() method.

9. Inside mul():

    b = 2

    c = 3

Multiplication:

b * c = 2 * 3 = 6

10. It displays:

Multiplication = 6

11. The statement

obj.add();

calls the add() method.

12. Addition is calculated:

b + c = 2 + 3 = 5

13. It displays:

Addition = 5

14. The statement

obj.a

accesses variable a from interface I1.

15. The value of a is 90.

16. It displays:

Value of a = 90

17. The statement

obj.x

accesses variable x from interface I2.

18. The value of x is 10.

19. It displays:

    Value of x = 10

20. Program execution ends.

Output

Multiplication = 6

Addition = 5

Value of a = 90

Value of x = 10

I1 Extends I2

```
package javacore;

interface I2 {
   void add();
}

interface I1 extends I2 {
   void mul();
}

class C1 implements I1 {

   public void add() {
      System.out.println("Addition = 5");
   }
```

```
    public void mul() {
        System.out.println("Multiplication = 6");
    }

    public static void main(String[] args) {

        C1 c = new C1();

        c.add();
        c.mul();
    }
}
```

Execution Steps

1. Start the program.

2. The main() method starts execution.

3. I1 extends I2.

   Therefore, I1 inherits the add() method from I2.

4. C1 implements I1.

   Therefore, C1 must provide implementations for:

      add()

      mul()

5. An object of C1 class is created:

      C1 c = new C1();

6. The statement

    c.add();

  calls the add() method of C1.

7. The add() method displays:

    Addition = 5

8. The statement

    c.mul();

  calls the mul() method of C1.

9. The mul() method displays:

    Multiplication = 6

10. Program execution ends.

Output

Addition = 5

Multiplication = 6

I1 Extends I2,I3

```
package javacore;

interface I2 {
   void add();
}

interface I3 {
   void sub();
```

```
}

interface I1 extends I2, I3 {
   void mul();
}

class C1 implements I1 {

   public void add() {
      System.out.println("Addition");
   }

   public void sub() {
      System.out.println("Subtraction");
   }

   public void mul() {
      System.out.println("Multiplication");
   }

   public static void main(String[] args) {

      C1 o = new C1();

      o.add();
      o.sub();
      o.mul();
   }
}
```

Execution Steps

1. Start the program.

2. The package "javacore" is loaded.

3. Interface I2 is defined.

   → It contains method add().

4. Interface I3 is defined.

   → It contains method sub().

5. Interface I1 extends both I2 and I3.

   → I1 inherits add() and sub().

   → I1 also contains mul().

6. Class C1 implements I1.

   → C1 must implement:

     add()

     sub()

     mul()

7. JVM searches for the main() method.

8. main() method starts execution.

9. Object of C1 is created:

     C1 o = new C1();

10. o.add() is called.

→ Prints "Addition"

11. o.sub() is called.

→ Prints "Subtraction"

12. o.mul() is called.

→ Prints "Multiplication"

13. Program execution ends.

Output

Addition

Subtraction

Multiplication